

**NMAM INSTITUTE OF TECHNOLOGY, NITTE***(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (Biotechnology) (Credit System) Degree Examinations**

January 2023

**19BTC114 – ENVIRONMENTAL BIOTECHNOLOGY**

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

| Unit – I   |                                                                                                                       | Marks | BT* | CO* | PO*       |
|------------|-----------------------------------------------------------------------------------------------------------------------|-------|-----|-----|-----------|
| 1.         | a) Explain the two pathways for biodegradation of phenolic compounds.                                                 | 10    | L*2 | 2   | 1,2       |
|            | b) What are the factors on which bioaccumulation depends?                                                             | 10    | L2  | 1   | 1         |
| 2.         | a) How are biosurfactants produced?                                                                                   | 10    | L2  | 2   | 1,2       |
|            | b) Explain 3 types of synthetic detergents with examples.                                                             | 10    | L2  | 2   | 1,2       |
| 3.         | a) How bioaccumulation of inorganic toxicants occurs? Explain in detail.                                              | 10    | L2  | 1   | 1,2       |
|            | b) What are polychlorinated biphenyls? List the properties of polychlorinated biphenyl's.                             | 10    | L2  | 2   | 1,2       |
| Unit – II  |                                                                                                                       |       |     |     |           |
| 4.         | a) Write the generalized flow diagram for waste water treatment.                                                      | 10    | L2  | 3   | 1         |
|            | b) Classify Biomedical waste with suitable examples.                                                                  | 10    | L2  | 4   | 1         |
| 5.         | a) Explain a scheme that is used for biological nitrogen removal.                                                     | 10    | L2  | 3   | 1,2,5,6,7 |
|            | b) What are the different types of membrane process that are used in waste water treatment and give example for each? | 10    | L2  | 3   | 1,2,5,6,7 |
| 6.         | a) What are the different membrane modules that are used in waste water treatment? Explain any two in detail.         | 10    | L2  | 3   | 1,2,5,6,7 |
|            | b) Explain in detail any 4 methods of land filling.                                                                   | 10    | L2  | 4   | 1         |
| Unit – III |                                                                                                                       |       |     |     |           |
| 7.         | a) With a neat flow sheet explain the step by step procedure to produce Bioethanol.                                   | 10    | L2  | 5   | 1,2,5,6,7 |
|            | b) Write a note on the steps that are involved in biofilm formation.                                                  | 10    | L2  | 5   | 1,5,6,7   |
| 8.         | a) What are the different types of Biofertilizers?                                                                    | 10    | L2  | 5   | 7         |
|            | b) Explain the methods of biomining in detail.                                                                        | 10    | L2  | 5   | 5         |

BT\* Bloom's Taxonomy, L\* Level; CO\* Course Outcome; PO\* Program Outcome

\*\*\*\*\*